

Problem 18.16

An estate provides a perpetuity with payments of X at the end of each year. Seth, Susan, and Lori share the perpetuity such that Seth receives the payments of X for the first n years and Susan receives the payments of X for the next m years, after which Lori receives all the remaining payments of X . Which of the following represents the difference between the present value of Seth's and Susan's payments using a constant rate of interest?

- (a) $X [a_{\overline{n}|} - v^n a_{\overline{m}|}]$
- (b) $X [\ddot{a}_{\overline{n}|} - v^n \ddot{a}_{\overline{m}|}]$
- (c) $X [a_{\overline{n}|} - v^{n+1} a_{\overline{m}|}]$
- (d) $X [a_{\overline{m}|} - v^{n-1} a_{\overline{m}|}]$
- (e) $X [va_{\overline{n}|} - v^{n+1} a_{\overline{m}|}]$

Solution

Seth receives the first n payments of X at the end of each year. Therefore, the present value of Seth's payments is

$$Xa_{\overline{n}|}.$$

Susan receives the next m payments of X , beginning after Seth's n payments. At time n , the value of Susan's m payments is

$$Xa_{\overline{m}|}.$$

Discounting this value back n years gives Susan's present value:

$$Xv^n a_{\overline{m}|}.$$

Thus, the difference between the present value of Seth's payments and Susan's payments is

$$Xa_{\overline{n}|} - Xv^n a_{\overline{m}|}.$$

Factoring out X ,

$$X [a_{\overline{n}|} - v^n a_{\overline{m}|}].$$

Therefore, the correct choice is

$$\boxed{(a)}.$$